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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

IS2212-1 – BUSINESS INTELLIGENCE AND ITS APPLICATIONS
(For IS) .

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is NOT a core function of a business enterprise organization?
A) Marketing
B) Finance
C) Production
D) Personal Hobbies
2. Information users in a business organization typically require
A) Constant Software Updates
B) Accurate and Timely Information
C) High-Level Programming skills
D) Open-source Platforms
3. What distinguishes structured data from semi-structured data?
A) Structured data is less organized than semi-structured data
B) Structured data does not use tags or markers, whereas semi-structured data does
C) Semi-structured data is organized into rows and columns, whereas structured data is not
D) Structured data is more flexible in format compared to semi-structured data
4. Which of the following best describes the data in a NoSQL database?
A) Structured
B) Unstructured
C) Semi-Structured
D) Data not stored digitally
5. Business Analytics is defined as
A) The use of historical data alone for decision-making
B) A systematic approach to interpreting data to forecast future trends and make data-driven decisions
C) An automated tool for generating reports
D) A database management system
6. How can Business Intelligence be utilized to analyze data for past, present, and future perspectives?
A) By examining historical performance, real-time operations, and predictive trends
B) By focusing solely on past performance reports
C) By analyzing only current operational data
D) By generating random data projections
7. Which of the following is an example of a Business Intelligence application?
A) Customer Relationship Management (CRM)
B) Enterprise Resource Planning (ERP)
C) Predictive Analytics Tool
D) Data Entry Software
8. Which of the following is NOT a typical component of a Business Intelligence (BI) framework?
A) Data Sources
B) Data Integration
C) Data Warehousing
D) Hardware Infrastructure
9. Which of the following is a key responsibility of a BI Developer?
A) Setting up and maintaining database servers
B) Creating and optimizing BI reports and dashboards
C) Managing network security
D) Handling customer service inquiries

10. Which component of a BI system is responsible for generating and distributing reports to end-users?
 A) Data Warehouse B) BI Reporting Tools
 C) Data Integration Layer D) Data Mining Tools
11. What is the main difference between a Data Mart and a Data Warehouse?
 A) A Data Mart is larger and more comprehensive than a Data Warehouse B) A Data Mart is focused on specific business areas, while a Data Warehouse is organization-wide
 C) A Data Mart manages real-time data, while a Data Warehouse handles historical data D) A Data Mart is used for daily operations, while a Data Warehouse is used for real-time analysis
12. Which technology is essential for integrating data from multiple sources into a single repository?
 A) Data Entry Tools B) ETL Technologies
 C) Data Visualization Software D) Database Management Systems
13. Which of the following is a primary objective of data integration?
 A) To simplify user interfaces B) To consolidate and harmonize data from different sources for comprehensive analysis
 C) To increase the speed of real-time transactions D) To create visual reports for stakeholders
14. Which schema type is best suited for a simple, denormalized data model with a single central fact table?
 A) Star Schema B) Snowflake Schema
 C) Galaxy Schema D) Data Mart Schema
15. What does the term "Dimensional Modeling" refer to?
 A) Modeling data for real-time transactional systems B) Organizing data into a format optimized for analytical and reporting queries
 C) Designing user interfaces for data entry D) Managing data security and access
16. In a data warehouse, what is typically stored in Dimension Tables?
 A) Measures and metrics B) Metadata and data lineage
 C) Raw transaction logs D) Descriptive attributes and context for facts
17. Which of the following is NOT a common type of data model?
 A) Conceptual Data Model B) Logical Data Model
 C) Physical Data Model D) Statistical Data Model
18. What is the primary purpose of using performance measures in a business?
 A) To manage daily operational tasks B) To evaluate and improve the effectiveness and efficiency of business processes
 C) To design marketing strategies D) To manage employee schedules
19. What is a Key Performance Indicator (KPI)?
 A) A tool used for managing project budgets B) A software application used for data visualization
 C) A specific, measurable value that indicates how effectively a company is achieving its key business objectives D) A method for collecting customer feedback
20. What is the benefit of using fact-based decision-making in an organization?
 A) It ensures decisions are made based on subjective opinions B) It provides a reliable basis for making informed and objective business decisions
 C) It limits the use of performance data in decision-making D) It focuses only on short-term operational issues

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Explain the different categories of users and the common expectations from Internet applications.

Marks BT* CO* PO*

8 L*2 1 1

b)	Consider an online electronics shop. Design a 1-D, 2-D and 3-D table for this shop. Illustrate slice, dice, roll up, drill down operation on the 3-D table that is designed.	8	L4	2	2
2. a)	Explain the differences between OLAP and OLTP.	8	L2	2	1
b)	Mention the three types of data. For an online garment shop Identify the possible types of data.	8	L2	2	1
3. a)	What is Unstructured data? With an example explain how inconsistent data formats causes problems in the integration of the data.	8	L2	2	1
b)	Consider a stationary shop database and write any four OLTP and four OLAP queries.	8	L3	2	2
Unit – II					
4. a)	Draw the BI component framework and explain the business layer	8	L2	3	1
b)	Consider the online shopping example and illustrate four poor quality data and method of correction.	8	L3	4	2
5. a)	Define BI. How is it different from ERP?	8	L2	3	1
b)	XYZ an online retail company, wants to improve its data analytics capabilities by designing a new data warehouse. The company needs to track and analyze sales transactions, customer behavior, product performance, and store locations. The goal is to create a star schema that will enable efficient querying and reporting for business insights. Design a Star scheme for this online retail store which supports the analysis of sales performance customer interactions and product information.	8	L3	4	2
6. a)	FinTech Solutions, a rapidly growing financial technology company, is undertaking a major renovation of its Customer Relationship Management (CRM) system to better manage customer interactions, streamline sales processes, and improve customer service. The company has recently hired a Business Requirement Analyst to lead the project. As the newly appointed Business Requirement Analyst for this CRM overhaul project, you are tasked with ensuring that the final system meets the business needs of FinTech Solutions. Explain at least 4 tasks of the Requirement Analyst to enhance the customer service.	8	L3	4	2
b)	Explain the issues that prompt the company to go for data warehousing.	8	L2	3	1
Unit – III					
7. a)	Explain the salient attributes of a good metric.	8	L2	5	1
b)	Explain the types of report layouts with an example.	8	L2	5	1
8. a)	What are KPIs? What KPIs of an ENGG college will be most relevant for a Pre-university completed candidate who is looking for various ENGG colleges for admission?	8	L3	5	2
b)	Explain the features of good reporting.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
